

P2: Teaching-Learning and Evaluation Plan
PROGRAMME: BCA (3rd Semester - GENERAL)
Teaching - Learning & Evaluation Plan

Course Information:

Course Code: **23BCA3C01**

Course Title: **Database Management Systems**

Credits Units: **03**

Contact Hours + Experiential Hours: **45+45**

L-T-P-E: **3-0-0-3**

IA: ESE Weightage- **50:50** Pass Marks (IA & ESE): **40 (ESE-Min.18)** Aggregate Pass Marks: **40**

ESE Question Paper Marks: **50**

Special Examination Fees: **NA**

Pre-requisite (if any):

1. Students must have the basic knowledge of data and database.

Course Facilitator (s): Dr.Santhalakshmi.M, Dr. Clara shanthi, Dr.Suma, Dr.Pratima, Prof.Yashaswini B.M.

At the end of the programme, students will be able to	
PO 1	Computational Knowledge: Understand and apply mathematical foundation, computing and domain knowledge for the conceptualization of computing models from defined problems.
PO 2	Problem Analysis: Ability to identify, critically analyze and formulate complex computing problems using fundamentals of computer science and application domains.
PO 3	Design / Development of Solutions: Ability to transform complex business scenarios and contemporary issues into problems, investigate, understand, and propose integrated solutions using emerging technologies.
PO 4	Conduct Investigations of Complex Computing Problems: Ability to devise and conduct experiments, interpret data and provide well informed conclusions.
PO 5	Modern Tool Usage: Ability to select modern computing tools, skills and techniques necessary for innovative software solutions.
PO 6	Professional Ethics: Ability to apply and commit professional ethics and cyber regulations in a global economic environment.
PO 7	Life-long Learning: Recognize the need for and develop the ability to engage in continuous learning as a Computing professional.
PO 8	Project Management: Ability to understand management and computing principles with computing knowledge to manage projects in multidisciplinary environments.

PO 9	Communication Efficacy: Communicate effectively with the computing community as well as society by being able to comprehend effective documentations and presentations.
PO 10	Societal & Environmental Concern: Ability to recognize economic, environmental, social, health, legal, ethical issues involved in the use of computer technology and other consequential responsibilities relevant to professional practice.
PO11	Individual & Team Work: Ability to work as a member or leader in diverse teams in multidisciplinary environment.
PO12	Innovation and Entrepreneurship: Identify opportunities, entrepreneurship vision and use of innovative ideas to create value and wealth for the betterment of the individual and society.
Program Specific Outcomes (PSO's)	
PSO1	Pertain current knowledge and adapting to emerging applications of Mathematics, Science fundamentals in the field of Computer Science Applications
PSO2	Exhibit proficiency in identifying, formulating and analysing complex problems in the computer environment.
PSO3	Ability to create, select and apply appropriate modern techniques for solving complex issues.
PSO4	Explore technical knowledge in diverse areas of Computer Applications and experience a conducive environment in nurturing skills for successful carrier and higher studies.

Course Objectives (COB's)

COB1	Implement knowledge to build database design.
COB2	Describe the concepts of normalization and transactions.
COB3	Adapting various SQL queries to operate database and to handle in different contexts.
COB4	Implement logical structure of the database using SQL.
COB5	Analyse the concepts of normalization and transactions.

Course Outcomes (CO's)

At the end of the course, students will be able to

S.No.	Course outcome	Description	Bloom's Taxonomy Level
1.	CO1	Articulate the various data models used in database design and Make use of ER diagram and normalization techniques	L3
2.	CO2	Implement the SQL operations on Database.	L3
3.	CO3	Summarizing the concept of normalization and transaction, concurrency control and recovery in databases.	L2
4.	CO4	Implement various SQL queries to perform various operations on databases.	L3
5.	CO5	Apply the database concepts, technology and create the relations by specifying primary and foreign keys	L3

CO-PO/PSO Mapping:

Course Outcome	Bloom's Taxonomy Levels	Programme Outcomes (PO's) Mapped												Program Specific Outcomes (PSO's)			
		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1 L3	PSO2 L3	PSO3 L2	PSO4 L6
CO1	L3	1	-	2	2	-	-	2	-	2	-	1	-	1	-	-	-
CO2	L3	1	2	2	2	-	-	2	-	2	-	1	-	1	-	-	-
CO3	L2	2	1	2	2	-	-	2	-	2	-	2	-	1	-	1	-
CO4	L3	2	1	2	2	-	-	2	-	2	-	2	-	1	-	-	-
CO5	L3	1	2	2	2	-	-	2	-	2	-	2	-	1	-	-	-

Total	7	6	10	10	-	-	10	-	10	-	8	-	5	-	1	-
Average	1.4	1.5	2	2	-	-	2	-	2	-	1.6	-	1	-	1	-

Course Contents:

Module	Details	Contact Hours
I	Purpose of Database System -- Views of data ,- Data Models – Database System Architecture – Database users and Administrator – Entity- Relationship model (E-R model) – E-R Diagrams -- cardinality relation, Types of cardinality, Introduction to relational databases, Dr EF Codd rules RELATIONAL MODEL: Normalization: Functional Dependencies – Non-loss Decomposition – Types of normalization, First, Second, Third Normal Forms, Dependency Preservation – Boyce/Codd Normal Form- Multivalued Dependencies and Fourth Normal Form	09
II	Database Languages –SQL fundamentals: Sub languages of SQL, DRL, DML, TCL, DDL, DCL. DRL Language, DRL: SELECT syntax, literal, Command, Logical Operator, Range Searching, Pattern Matching, order by clause. Group By Clause, Built-in Functions in SQL: Single row multiple row, Number, character, date, conversion function, general functions Grouping/aggregate functions. Grouping data from Tables in SQL,	09
III	Joins, Joining Multiple Tables, different types of joins, Sub queries, Types of subqueries, single row subquery, multiple row subquery, Top N Analysis, DML Statements insert, update, delete. Merge. Oracle data types, TCL Command: commit, rollback and savepoint Set Operators: Union, intersect & Minus Clause	09
IV	DDL commands (CREATE, drop, alter, rename, truncate) Data Types, TABLE Command, Constraints, primary key, foreign key, check, not null, unique, setting constraints using Column level & table Level	09

	Constraints, defining different constraints on the table, Defining Integrity Constraints in the ALTER Views: Create view, types of view, operations on views. dictionaries for table, view, user, constraints etc, creating view, Renaming the Column of a view, Updating, Selection, destroying view, Creating Indexes,	
V	DCL: Granting Permissions, Revoke permission - Creating and managing User, Create, Role, provide privileges, dictionaries, Triggers syntax, Enabling/Disabling, Triggers; Transaction Concepts - Transaction Recovery - ACID Properties, Serializability, types of serializability - conflict serializability, view serializability.	09

Text Books (TB):

Text Book:	
1.	Abraham Silberschatz, Henry F. Korth, S. Sudharshan, "Database System Concepts", Fifth Edition, Tata McGraw Hill, 2016
Reference books:	
1.	RamezElmasri, Shamkant B. Navathe, "Fundamentals of Database Systems", Fourth Edition, Pearson/Addison Wesley, 2017.
2.	Raghu Ramakrishnan, "Database Management Systems", Third Edition, McGraw Hill, 2013.
3.	The Complete Reference SQL Third Edition, James R. Groff, Mc Graw Hill, ISBN: 9781259003882
4.	SQL in a Nutshell, 4th Edition, Kevin Kline, Regina O. Obe, Leo S. Hsu, June 2022, O'Reilly Media, Inc., ISBN: 9781492088868

OR-1	https://www.educba.com/data-science/data-science-tutorials/oracle-tutorial/
OR-2	https://www.tutorialspoint.com/ms_sql_server/index.htm
OR-3	https://www.postgresqltutorial.com/
OR-4	https://www.khanacademy.org/computing/computer-programming/sql

Relevant Websites (RW):

RW-1	https://beginnersbook.com › RDBMS Concepts
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RW-2	https://www.tutorialspoint.com › sql › sql-rdbms-concepts
RW-3	https://education.oracle.com/de/oracle-database-12c-administrator-certified-associate/trackp_248
RW-4	https://www.javatpoint.com › dbms-normalization
RW-5	https://livesql.oracle.com/
RW-6	https://www.w3schools.com/mysql/mysql_rdbms.asp
RW-7	https://education.oracle.com/de/oracle-database-12c-administrator-certified-associate/trackp_248

Video Links (VL):

VL-1	https://www.youtube.com/watch?v=4qFZ-5i4GS8
VL-2	https://www.youtube.com/watch?v=73Z3-ekxS7Q
VL-3	https://www.youtube.com/watch?v=L5lhh-IM-m4
VL-4	https://www.youtube.com/watch?v=fvGwkJ8SA8g
VL-5	https://www.youtube.com/watch?v=pAB-z0knPpk
VL-6	https://www.youtube.com/watch?v=0OQJDd3QqQM
VL-7	https://www.youtube.com/watch?v=HAAhn--tZV8

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Sr.No.	Platform	Topic	CO	Link	Duration
MC-1	Linkedin	Learning Oracle Database 19 C	CO1,CO2,CO3,CO5	https://www.linkedin.com/learning/learning-oracle-database-19c/what-you-should-know?autoSkip=true&autoplay=true&resume=false&u=92695330	2 hours
MC-2	Linkedin	PostgreSQL Essential Training	CO1,CO2,CO5	https://www.linkedin.com/learning/postgresql-essential-training/manage-relational-data-with-postgresql?autoplay=true&u=92695330	2 hours 43 mins
MC-3	Linkedin	MicrosoftSQL Server Installation	CO2,CO4	https://www.linkedin.com/learning/microsoft-sql-server-2016-installation-and-administration/welcome?autoplay=true&u=92695330	2 hours 35 mins

MC-4	Linkedin	SQL Essential training	CO1,CO3	https://www.linkedin.com/learning/sql-essential-training-3/understanding-sql?autoplay=true&u=92695330	3 hours
MC-5	Linkedin	Database Foundations	CO1,CO5	https://www.linkedin.com/learning/database-foundations-administration-8375189/administer-a-database-server?u=92695330	2 hours 7 mins
MC-6	Linkedin	Oracle Database 19c: Basic SQL	CO3	https://www.linkedin.com/learning/oracle-database-19c-basic-sql	1 hour
Total MOOC Course integration with Certification					13 hours

Abbreviations & Expansions					
Pedagogy/Activity planned			Mode of Delivery		
P1	PPT presentation & Classroom interaction		M1	Synchronous – PPT	
P2	Blended Learning with Hands on		M2	Synchronous - Discussion	
P3	Flip Class & Module Quiz				
A1	Activity-1				
A2	Activity-2				

S.NO	Activities	Platform
1	Activity 1	Online Certification Course (Min 12 hours)
2	Activity 2	Mini Project

Session-wise Planning:

Module	Session	Topic	PPT Link	Readings and References	Pedagogy/Activity Planned	CO	Mode of delivery
I	1	TLEP discussion, Introduction, Purpose of Database System	https://docs.google.com/presentation/d/1H8KJU4dAa-gHaHAv6iHV3ObwfzLr95/edit?usp=drive_link&ouid=116018608028545558107&rtopof=true&sd=true	1.TB1 2.RB1,RB2 3.RW1	P1	CO1	M1

	2	Data Models ,Database users and administrator	https://docs.google.com/presentation/d/1JlkvCjDXERMIjIRwpXf-JA-nJQyJkO8i/edit?usp=drive_link&ouid=116018608028545558107&rtpof=true&sd=true		P1		M1
	3	Database System Architecture	https://docs.google.com/presentation/d/164T_MBv2MC-54ihkPSYKM2s4iWsLdpWD/edit?usp=drive_link&ouid=116018608028545558107&rtpof=true&sd=true		P1		M1
	4	Entity– Relationship model (E- R model)	https://docs.google.com/presentation/d/1isQG651otRm2IKPpFaT2qxOpn-xonyo/edit?usp=drive_link&ouid=116018608028545558107&rtpof=true&sd=true		P1		M1,M2
	5	cardinality relation, Types of cardinality	https://docs.google.com/presentation/d/1tpuDDMqOsl-ZQ7XWxAYWA2J6h_JiyJrS/edit?usp=drive_link&ouid=116018608028545558107&rtpof=true&sd=true		P1		M1,M2

	6	Introduction to relational databases, Dr EF Codd rules	https://docs.google.com/presentation/d/1gkjiF-RNaULB8cHEd1BnE8A-BoWFxmXk/edit?usp=drive_link&ouid=116018608028545558107&rtpof=true&sd=true		P1		M1
	7	Normalization: Functional Dependencies – Non-loss Decomposition – Types of normalization, First	https://docs.google.com/presentation/d/1KKEWrjLTzSM-uFV6VLHXoXLB7CAv490v/edit?usp=drive_link&ouid=116018608028545558107&rtpof=true&sd=true		P1		M1
	8	Second, Third Normal Forms, Dependency Preservation	https://docs.google.com/presentation/d/1RLVxvgluU2rI6nvfDybxocYu4JrKH3GL/edit?usp=drive_link&ouid=116018608028545558107&rtpof=true&sd=true		P1		M1
	9	Boyce/Codd Normal Form- Multivalued Dependencies and Fourth Normal Form	https://docs.google.com/presentation/d/1qAdzM45va_N6wRzDjliBFUXKooPi-A7_/edit?usp=drive_link&ouid=116018608028545558107&rtpof=true&sd=true		P1		M1

II	1	TLEP discussion, SQL Fundamentals, - Downloading SQL Version - 32-bit versus 64-bit - Installing	https://docs.google.com/presentation/d/197B3RBMnlGdH0s74WW0P4xZvXzMzPWZN/edit#slide=id.p1	1. TB1 2. https://db-book.com/slides-dir/index.html 3. RW1, RW2, RW3,	PPT Presentation & Classroom interaction	CO2	M1, M2
	2	Sub languages SQL, SQL Commands, Data types in SQL.	https://docs.google.com/presentation/d/1DwRZT1HTKbTkjezQsuB2obcutSfL9Uc5/edit#slide=id.p1	https://www.youtube.com/watch?v=ibOzwFRm32w&list=PLiLpmqwkwkCt0QeXD8j7BwIoOaBGBRrZC	PPT Presentation & Classroom interaction		M1
	3	SQL Commands with short examples, DDL, DML, DCL, TCL, DQL(DRL)	https://docs.google.com/presentation/d/15YZOCIKvVn00MYUz1oNmlojhxCRXa9Qn/edit#slide=id.p1	https://www.youtube.com/watch?v=4qFZ-5i4GS8	PPT Presentation & Classroom interaction		M1

	4	DRL:SELECT syntax,literal, Command	https://docs.google.com/presentation/d/16-z91DbuTrKBB AZCPhUem2YhtTf GTuTh/edit#slide=id.p1		PPT Presentat ion & Classroo m interactio n		M1
	5	SQL Operators, SQL Tables & operations on Table	https://docs.google.com/presentation/d/1A-HyKCdFYPrsI YxFWEOcnOiw7t E4oWK/edit#slide=id.p1		PPT Presentat ion & Classroo m interactio n		M1
	6	Range Searching, Pattern Matching, order by clause. Group By Clause intro..	https://docs.google.com/presentation/d/1y_0i24oyl3cyQm5 xkv7VBQgYjQMT LbhJ/edit#slide=id.p1		PPT Presentat ion & Classroo m interactio n		M1
	7	Built-in Functions in SQL: Introduction, Single row multiple row	https://docs.google.com/presentation/d/1fURZ07r9l1r64Ys ulobi0JyHtNIZMm D-/edit#slide=id.p1		PPT Presentat ion & Classroo m interactio n		M2

	8	Number, character, date, conversion function, general functions Grouping/aggregate functions.	https://docs.google.com/presentation/d/1RhhvSovGbAhMGBtX9eooZSHq8BPHfoiF/edit#slide=id.p1		PPT Presentation & Classroom interaction		M1
	9	Grouping data from Tables in SQL	https://docs.google.com/presentation/d/1G_7VzsDXUQYIEF6j7V_nL_CmNov9zioL/edit#slide=id.p1		PPT Presentation & Classroom interaction		M1
III	1	Joins, Joining Multiple Tables	https://docs.google.com/presentation/d/1-yfVoQGXC9nmgasKVr-QusHHYonpZ38EoUdEbY4Lvw/edit#slide=id.p1	TB1, RB1, RB2	P1, P2		M1
	2	Different types of Joins	https://docs.google.com/presentation/d/1q8crAmEFRsTR9tMFrh2WchRKPsE80nMysEw4UTmODOQ/edit#slide=id.p1		P1, P2		M1
	3	Different types of Joins Continued	https://docs.google.com/presentation/d/1poldzD4JA6BNzddnpKl8jXviNr0i3l3RGjPRe16PGyQ/edit#slide=id.p1		P1, P2		M1

	4	SubQueries, Types of subqueries, single row subquery, Multiple row subquery	https://docs.google.com/presentation/d/1pOFM83rPjD57btYiEOWaTrr-LWuxPalIbsM4EbF7Jk4/edit#slide=id.p1		P1,P2		M1
	5	DML statements Insert,, Update	https://docs.google.com/presentation/d/17zKA52rbXctcdJKFgRfeFTqgHaOG-KJpqCgnnL3mtCA/edit#slide=id.p1		P1,P2		M1
	6	DML statements Delete, Merge	https://docs.google.com/presentation/d/1QbJQyx7BuUECh_a5EHbZv_qUg1FuLKCmWT80BH_KfSMM/edit#slide=id.p1		P1,P2		M1
	7	Top N analysis, Oracle Data Types	https://docs.google.com/presentation/d/1ULmCK1HAwdk4jVQPvtXPuAS4kF_fV1CsEu64rKRt0Hk/edit#slide=id.p1		P1,P1		M1
	8	TCL commands Commit, Rollback, Savepoint	https://docs.google.com/presentation/d/1DyZ6QdAJfv2BWy4dBF6XFmdpT-fwggOp7imhYRCrsr0/edit#slide=id.p1		P1,P2		M1

	9	SET operators UNION,MIN US and INTERSECT	https://docs.google.com/presentation/d/1vZ0i8a2S7vAMVjW8sfsWcldCeBvAvBvWG1i6W2LNJCk/edit#slide=id.p1		P1,P2,P3		M1
IV	1	Creating and managing tables,naming rules,Queryin g data dictionary	https://docs.google.com/presentation/d/17pQhVp6NXpJtAN6eV-FiRNINwkoRK4O9/edit?usp=sharing&ouid=112601164060256616862&rtpof=true&sd=true	TB1 RW7	P1,P2	CO4	M1
	2	Data types,date time data types,creating table by using sub query	https://docs.google.com/presentation/d/1BN86YoFWa45LGhY8xUWg8LqrDyoluOxg/edit?usp=sharing&ouid=112601164060256616862&rtpof=true&sd=true		P1,P2		M1
	3	Alter table,Drop table,truncate table	https://docs.google.com/presentation/d/1JAacba5L0kLJA0dqBeP11vsklRRepTRZ/edit?usp=sharing&ouid=112601164060256616862&rtpof=true&sd=true		P1,P2		M1

	4	Constraints-Not null,unique,Primary key constraints	https://docs.google.com/presentation/d/1yMS1LzdQhzbW2ATpq8whXZMB0s8tMHJH/edit?usp=sharing&ouid=112601164060256616862&rtpof=true&sd=true		P1,P2		M1
	5	Foreign key,check constraint,adding,dropping enabling constraint	https://docs.google.com/presentation/d/1Z4Be7A_4plzyJh8UooL9T3krMX9CVyBd/edit?usp=sharing&ouid=112601164060256616862&rtpof=true&sd=true		P1,P2		M1
	6	cascading constraint,viewing constraint	https://docs.google.com/presentation/d/1RjD64Diqyf0kU9o9RQSyvimsqHgbLXnc/edit?usp=sharing&ouid=112601164060256616862&rtpof=true&sd=true		P1,P2		M1
	7	what is view,creating view	https://docs.google.com/presentation/d/1Ao8p_2jLRWE7dlt6-7DmG2QW0_ozfX2X/edit?usp=sharing&ouid=112601164060256616862&rtpof=true&sd=true		P1,P2		M1

	8	retrieving data from view, querying, modifying, creating complex view, rules for performing DML operations	https://docs.google.com/presentation/d/195Da5_dlszG6ju5apsihe1EgLUbEEOs4/edit?usp=sharing&ouid=112601164060256616862&rtpof=true&sd=true		P1,P2		M1
	9	with check option clause, removing view, inline view, Top N analysis	https://docs.google.com/presentation/d/1CNZAvSYafMH8XUr7D3tu8WptWYNVtbiY/edit?usp=sharing&ouid=112601164060256616862&rtpof=true&sd=true		P1,P2		M1
V	1	DCL overview & commands for	https://docs.google.com/presentation/d/1kMA3OPKoKIE6zr2fVT_jXXTSey8Umga/edit?usp=sharing&ouid=112601164060256616862&rtpof=true&sd=true	TB1,RB1,RB2	P1,P2		M1
	2	grant, Revoke permission, Difference between DCL & TCL, User privileges intro	https://docs.google.com/presentation/d/1p8A_WLgqvPjPMf8MAmdviFDp0FtWj8Ou/edit?usp=sharing&ouid=112601164060256616862&rtpof=true&sd=true		P1,P2		M1

	3	Triggers	https://docs.google.com/presentation/d/1hsqG7vrUdXowiP1-C2c6IJrO_f9dH8yt/edit?usp=sharing&ouid=112601164060256616862&rtpof=true&sd=true		P1,P2		M1
	4	Creating,Managing the user,provide privileges	https://docs.google.com/presentation/d/10ANoHRdoCJKS_eZe44JDBejbkzvQfgmzNdscq7PgiZs0/edit#slide=id.p1		P1,P2		M1
	5	Creating,Managing the user,provide privileges continued	https://docs.google.com/presentation/d/1_KlGbhuC791y0kFbTsp4pK4OAIId12tMwOFN-47-lq-c/edit#slide=id.p1		P1,P2		M1
	6	Transaction Concepts,Transaction recovery-ACID properties	https://docs.google.com/presentation/d/1yoF4I0jGmrXcNTxwup39ktWL6z0dEyCCN9zwXkzZyGY/edit#slide=id.p1		P1,P2		M1
	7	Serializability, types of serializability	https://docs.google.com/presentation/d/1HMTjOqfLKUD84vkY6Wi4YwI7Ecel6lfu/edit#slide=id.p1		P1,P2		M1

	8	conflict serializability	https://docs.google.com/presentation/d/1-N6I7OTpPAph2_Zulh1efzdPy_Cy3Do7/edit#slide=id.p1		P1,P2		M1
	9	view serializability.	https://docs.google.com/presentation/d/1czPR0CcTozkBF LbxhWCaWXTy2d IGMnZ3/edit#slide=id.p1		P1,P2		M1

Assessment Scheme: IA: ESE – 50:50

Sl. No.	Assessment Instrument	Formative/ Summative	Frequency	Marks	CO
1.	Activity-1 – LinkedIn	Formative	1	15	CO1, CO2, CO3, CO4, CO5
2.	Activity-2 – Coursera Course		1	15	CO3, CO4, CO5
3.	Class participation	Formative	Continuous	05	CO1, CO2, CO3, CO4, CO5
4.	Internal Test 1	Formative	1	7.5	CO1, CO2, CO3
5.	Internal Test 2	Formative	1	7.5	CO3, CO4, CO5
5.	End Semester Examination	Summative	1	50	CO1, CO2, CO3, CO4, CO5
Total				100	

Bloom's Taxonomy-Revised

LEVEL	DESCRIPTION	MEANING	ACTION VERBS
6	Creating	Can the student create a new product or POV?	Assemble, construct, create, change, combine, compose, design, develop, formulate, invent, modify, organize, propose, theorize, write
5	Evaluating	Can the student justify a stand or decision?	Appraise, agree, assess, argue, conclude, decide, defend, judge, prioritize, prove, rate, recommend, select, support, value
4	Analyzing	Can the student distinguish between different parts?	Contrast, compare, criticize, differentiate, discriminate, dissect, distinguish, examine, experiment, operate, question, simplify, test
3	Applying	Can the student use information in a new way?	Choose, demonstrate, dramatize, employ, illustrate, interpret, schedule, sketch, solve, use
2	Understanding	Can the student explain ideas and concepts?	Classify, describe, discuss, explain, identify, infer, locate, outline, paraphrase, recognize, report, summarize, select, translate
1	Remembering	Can the student recall or remember information?	Define, duplicate, find, list, label, match, memorize, name, omit, recall, repeat, state, spell, tell

Signature of the Faculty

Signature of the HoD